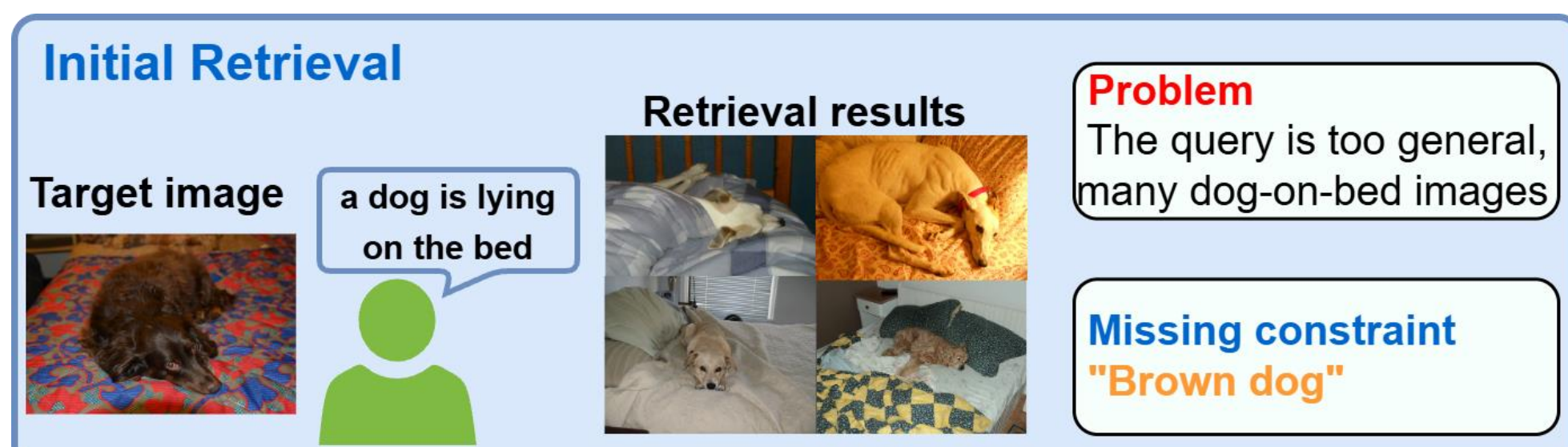


1 Motivation / Problems

- **Image retrieval** ranks database images according to a **user text query**.
- **Conventional retrieval** usually returns results in a **single step**.
- **Incomplete or ambiguous queries** can retrieve **visually similar distractors**.



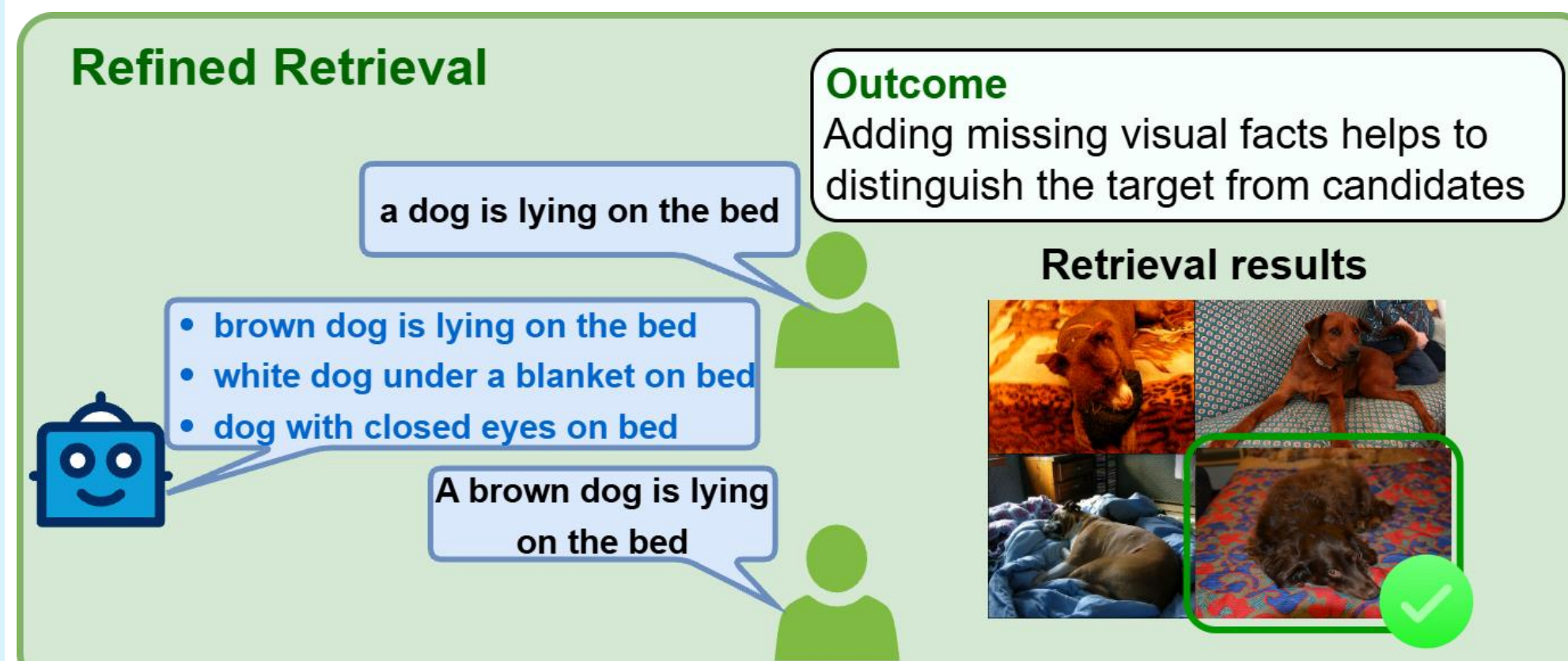
! The true target may not be ranked high due to missing details in the query.

2 Goal & Technical Challenges**Goal:**

Help users refine **ambiguous image-search queries** through **interaction**.

Challenges:

- Identify **missing visual details** in the current query.
- Select **query-relevant** and **discriminative cues** from retrieved candidates.
- Convert selected cues into **refinement suggestions**.

**3 Proposed Idea**

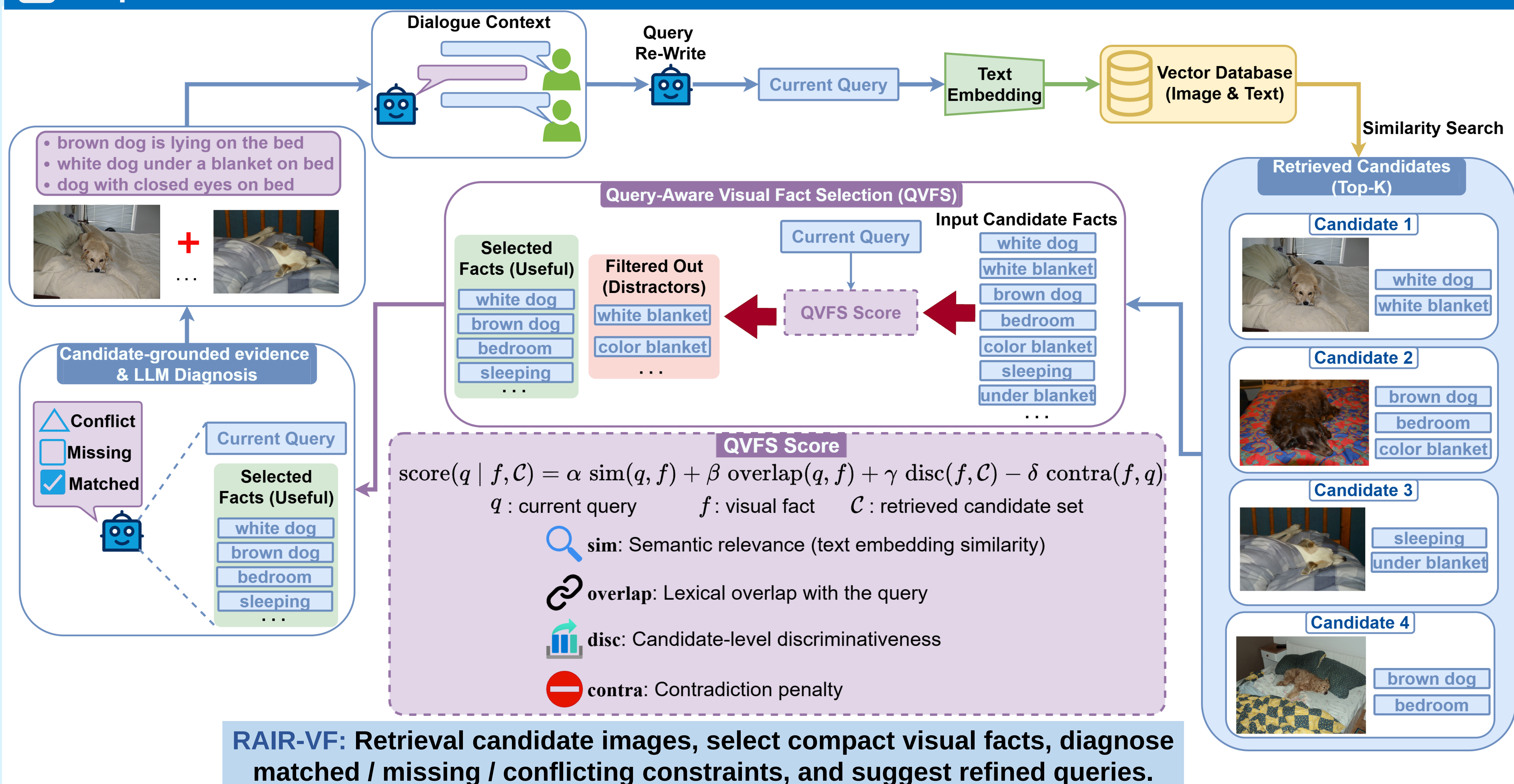
- Use **retrieved candidates** as evidence for **query refinement**.
- **Visual facts** are short text cues of visible details, e.g., "**brown dog**" or "**lying on bed**".
- **Query-aware Visual Fact Selection (QVFS)** selects **relevant** and **discriminative** facts.
- The **LLM** suggests the **next query refinement** from the selected facts.

Evidence Selection & Diagnosis**QVFS-selected visual facts**

- brown dog is lying on the bed
- white dog under a blanket on bed
- dog with closed eyes on bed

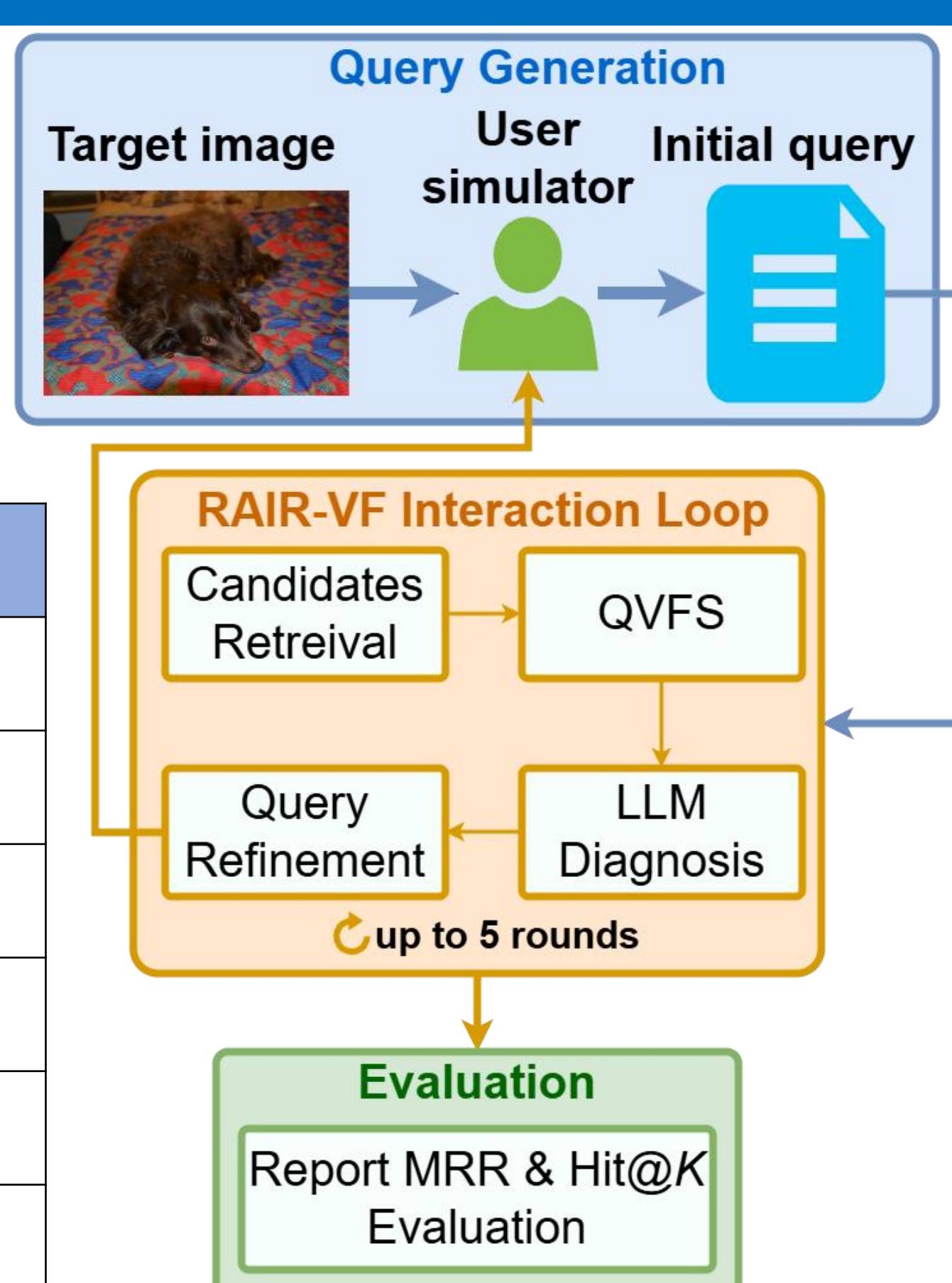
LLM Diagnosis

- **Matched:** dog, lying, bed
- **Missing:** brown dog, white dog
- **Distraction:** blanket color

4 Proposed Framework: RAIR-VF**5 Experiments & Results****Dataset:** Visual Dialog (Visdial) [1]**Metric:** Hit@K**Local LLM:** Gemma3-12B [2]**User simulator:** Qwen/Qwen2.5-VL-7B-Instruct [3]**Database:** PostgreSQL, openai/clip-vit-base-patch32 [4]**Similarity Search:** FAISS [5]

Turn	Hit@10	Hit@20	Hit@50
0	0.4444	0.5456	0.6769
1	0.4788	0.5794	0.7125
2	0.4913	0.6394	0.7856
3	0.5320	0.6931	0.8212
4	0.5969	0.7450	0.8531
5	0.6350	0.7756	0.8731

Retrieval improves consistently across interaction turns, with **Hit@10**, **Hit@20**, and **Hit@50** increasing by **+19.06**, **+23.00**, and **+19.54** pe+19.06, +23.00, and +19.62 percentage points, respectively.

**6 Conclusion & Future Works****Conclusion**

- **Candidate-grounded visual facts** support query refinement.
- **QVFS** provides compact and relevant evidence.
- **LLM diagnosis** helps clarify missing or conflicting constraints.
- **Multi-turn interaction** improves best-so-far retrieval performance.

Future Works:

- Extend **RAIR-VF** to larger datasets and more diverse interactive retrieval scenarios.
- Conduct **human-user studies** to evaluate the usefulness and interpretability of refinement suggestions.

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[2] Gemma Team, A. Kamath, J. Ferret, S. Pathak, et al., "Gemma 3 Technical Report," arXiv:2503.19786, 2025.

[3] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, et al., "Qwen2.5-VL Technical Report," arXiv:2502.13923, 2025.

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, et al., "Learning Transferable Visual Models From Natural Language Supervision," Proceedings of the International Conference on Machine Learning (ICML), 2021.

[5] J. Johnson, M. Douze, and H. Jegou, "Billion-scale Similarity Search with GPUs," IEEE Transactions on Big Data, 2019.